SAMPLE

Great Observatories: ALMA

The Atacama Large Millimeter Array

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Introduction

On the Chajnantor Plateau of northern Chile, five kilometers above sea level, sixty-six white antennas stand in the driest air on Earth and listen. The Atacama Large Millimeter/submillimeter Array, ALMA, is the largest ground-based astronomy project ever built, a machine the size of a small city assembled at an altitude where unacclimated visitors can barely walk, and its subject is the cold universe: the gas between the stars, the dust from which planets are made, the light of galaxies so distant that its wavelengths have been stretched a hundredfold on their way here. This book is the story of that machine, how it came to be built on one of the least forgiving landscapes a big project has ever chosen, what it has heard since its first antennas were linked, and why the coldest light in the sky took the most expensive radio receiver ever made to catch it.

The book follows the pattern of its series. The opening chapters trace the discovery and construction history: the three rival millimeter-array projects on three continents that merged into one, the diplomacy of a partnership spanning North America, Europe, and East Asia, the engineering of antennas built to work while flexing in millimeter precision, and the decade of construction that ended with the array's formal inauguration in 2013. The middle chapters explain the science and the instrument: why millimeter and submillimeter light hold the keys to star formation and the early universe, how interferometry turns a plateau of dishes into a single synthetic telescope, what the receivers and correlator actually do, and how astronomers compete for and use the array's time. The later chapters take up the discoveries, the standing challenges, the future of the instrument, and the curiosities of working at five thousand meters. A conclusion gathers the threads. Throughout, the book keeps the observer's-eye habit of its series, describing not only what the instrument does but what it is like to stand on the plateau beside it.

The book also keeps, as its series does, an account of the people: the engineers who assembled the antennas in the desert, the operators who run the driest nights, the drivers of the great transporters, and the medical and support staff who make a five-thousand-meter worksite habitable, because a machine this ambitious is as much an operation as an instrument, and the operation is part of the story. One promise, and one caution. The promise: the array is young enough that its results are arriving monthly, and the reader who finishes this book will know how to read each new result as it comes, because the instrument's logic is simpler than its cost. The caution: a book about a working observatory is a snapshot, and the numbers here, the proposal pressure, the wavelength bands, the known records, describe the array at a moment; ALMA will improve past this book, and the reader should expect it to.

The book's method is the method of its series, and a word on it belongs at the front. Every chapter answers a standing question: how the machine came to be, what it is, what it measures, what it has found, and what it will do next, and the answers are written to be read without a physics degree, in plain language, with the technical terms introduced as they are needed and the numbers given where they earn their place. The chapters also tell the story honestly, which means the cost overruns, the weather, the altitude's toll, and the politics of a three-continent partnership appear alongside the rings of the disks and the black holes of the sky, because an instrument's story without its difficulties is an advertisement rather than a book. What follows is the story of the machine at the dry apex of the Atacama: the biggest ear on the cold sky, and the people who keep it listening.

Chapter 1: The Discovery of ALMA

The subject of this chapter is not the discovery of the array's site but the arrival of the idea that built it, and the idea has a history worth telling in order. Millimeter astronomy began in the 1960s and 1970s as a frontier discipline, pursued by small groups with single dishes, because the atmosphere is nearly opaque at millimeter and submillimeter wavelengths and only the driest, highest sites allow the light through. The rewards drove the field forward anyway: millimeter light carries the spectral lines of carbon monoxide, the tracer of the molecular gas from which stars form, and of dozens of other molecules that chart the chemistry of the space between the stars; submillimeter light carries the glow of cold dust, the stuff of collapsing clouds and planet formation, and the redshifted glow of the earliest galaxies. By the 1980s each of the world's three radio-astronomy communities had drawn the same conclusion: the next instrument in this science had to be an array, many dishes linked as one, on a high dry site, and each community drew the plans separately.

North America's National Radio Astronomy Observatory designed the Millimeter Array, a project of the early 1990s. Europe, through the European Southern Observatory and its member institutes, pursued a plan for a Large Southern Array, larger and at a higher frequency limit. Japan's National Astronomical Observatory developed its own Large Millimeter Submillimeter Array concept, with emphasis on the shortest wavelengths and on

compact-array performance for wide-field imaging. Three projects, one site requirement, one decade of studies, and a realization that no single community could fund the full ambition: the mergers that followed are among the most consequential negotiations in the history of ground-based astronomy, and the array that resulted is the rare case of a project that is genuinely international not in its staff but in its capital, its design, and its management.

The site question was settled by measurement. Site testing through the 1990s, with the candidate Chajnantor Plateau in Chile's Atacama Desert monitored alongside alternatives in the American Southwest, Mexico, and elsewhere, found in the Chilean plateau a combination no rival matched: an altitude of five thousand meters, an atmosphere above the site so dry that the submillimeter transmission windows open for long stretches of the year, and a plateau floor flat enough and wide enough, some ten kilometers across at its working stretch, to hold an array spread across baselines of many kilometers. The tests involved years of radiometers, radiosondes, and opacity meters working through the site's seasons, and the Chajnantor data became the engineering foundation on which the array's receivers were designed. The plateau's name is said to mean place of departure in the language of the region's first peoples, a name the astronomers found fitting for a machine built to watch the departure of light from the universe's beginning.

With the merger agreed in principle by the partner organizations in the late 1990s, and the site campaign converging on Chile, the project's shape stabilized into a plan for dozens of high-precision antennas, a giant correlator to combine their signals, receivers spanning the full millimeter and submillimeter bands then reachable from the ground, and an array configuration that could be rearranged, antennas physically moved between concrete pads, to change its resolution from wide-field to ultra-sharp. The formal ground-breaking on the mountain came early in the 2000s, and the construction decade that followed is the subject of the chapter's closing pages: the first antennas arrived in 2007 and 2008, the first pair was linked into an interferometric fringe in 2009, the first observing programs of the early-science phase ran in 2011 to a proposal demand roughly nine times the available time, and the completed array was formally inaugurated in March 2013 in a ceremony held, as much of the project's life is held, at the lower support campus rather than on the thin air of the plateau itself. The construction left behind an organization of several hundred staff, two custom-built antenna transporters, and a working instrument that within its first full decade would take the images of planet formation that astronomy had awaited for half a century.

The construction decade itself deserves the chapter's closest look, because the machine was assembled under conditions that would have daunted a military engineering corps. The antennas were built in three lots by manufacturers on three continents, European, North American, and Japanese, to a common specification so exacting that the three lots perform as one instrument: a surface accuracy good to a fraction of the shortest wavelength observed, a pointing stability that holds the beam on a target through wind and thermal cycling, and an interface, the bolt pattern, the electronics, and the software, that lets any transporter carry any antenna to any pad. The first antennas reached Chile late in the 2000s, shipped in pieces to the coast, trucked to the support campus, assembled and tested at twenty-nine hundred meters, and only then carried up to the plateau; the first fringes between linked antennas came in 2009, and the early-science era began in 2011 with a partial array whose time was oversubscribed roughly ninefold, a figure that told the partners everything they needed to know about the demand waiting for the finished machine. The inauguration, in March of 2013, marked the formal end of construction, and the array has operated as a user facility ever since, its sixty-six dishes cycling through configurations the way a factory cycles shifts.

The early history also carries the project's own chronology of setbacks, and the honest account keeps them. The cost of the array outran the original estimates, as costs at such projects do, and the partnership's solution, a re-baselining that descoped the deliverables and phased the completion, is a case study in how a big science project survives its own budget; the global financial troubles of the era arrived mid-construction, and the array's funding survived them partly because the partnership's interdependence made withdrawal by any partner a treaty problem rather than a line-item decision. The high-altitude logistics wrote their own chapters: roads raised and rebuilt, a power line across the desert, a technical building built to work in air that punishes diesel engines and cooling systems alike, and a medical program whose staffing reflects the hardest worksite in professional astronomy. The finished machine cost, by the published figures of the partnership, more than a billion dollars, the most expensive ground-based astronomical instrument ever assembled, and the construction history is in that number's shadow: every hundred-ton antenna on the plateau is also a hundred-ton monument to a decade of project management conducted at the edge of the atmosphere.

One more strand of the early history deserves its paragraph, for it is the quiet foundation of everything after it. The

Chajnantor campaigns of the 1990s ran radiosondes, radiometers, and test receivers through years of the plateau's seasons, measuring the atmospheric transmission at the frequencies the planned receivers would use, and the data those campaigns produced, opacity against season, water vapor against hour, transmission against elevation, became the specification the antennas, the receivers, and even the observatory's calendar were built against. The site testing also left a legacy the field still uses: the measurement techniques pioneered for the Chajnantor decision are the standard instruments of every high-site evaluation since, and the plateau, by being studied first and hardest, became the calibration of mountain astronomy. The reader should file this fact beside the machine's headline numbers: the array is not only built on the best site the testers could find, it is built on the first site that was ever measured this way, the place where the discipline of choosing a mountain was invented.